Toward a four-dimensional Yang–Mills construction

Notebook 13: a uniform nonlinear constraint chart

Research working document

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Abstract

The complete corrected averaging map, iterated through fixed-size blocks, admits a complex neighborhood whose physical radius does not shrink with refinement. We prove this from its exact local coordinate, the contraction of its flat linear part and a geometric sum of physical scale factors. Its differential is uniformly close to the flat average in bounded-field and physical Hilbert norms. The flat minimizing extension then gives an exact analytic chart of the nonlinear constraint, including an exponentially local right inverse for its ordinary-coordinate differential. These are fixed-reference statements; the moving-background operator and the nonlinear action minimizer require separate identification.

Status. Research proofs checked by separate internal derivations and audits. No external certification or novelty claim is made. Source matching is to the pinned deposited transcription identified below.

1 A uniform analytic neighborhood for the corrected nonlinear average

Fix an odd one-step ratio $L \geq 3$, a refinement $M = L^j$, fine spacing $\eta = M^{-1}$ and final physical scale one. The reference background is the identity at every stage. Constants may depend on L, dimension, color norm and the specified averaging prescription, but not on j, M or compatible periodic volume. Pairings and Hilbert norms use physical lattice volume. All complex analytic maps below use the fixed flat-reference logarithmic coordinate. Their differentials at a nonzero field are not identified here with a separately left-trivialized nonflat source operator.

1.1 The exact source coordinate and the fixed local constants

Deposited Raw I:24320–24325 composes the corrected nonlinear operation, with its averaged reference background. Since that reference is initially the identity, it remains the identity. I:24344–24353 gives the logarithmic coordinate, its linear part and the quadratic remainder; I:24357–24369 identifies the flat linearization. The coordinate bridge I:47076–47087, using the maps defined at I:46965–46973, gives

$$F_L(X;c) = \frac{1}{i}\log\left[e^{-i\alpha_{c_-}(X)}e^{i\phi(X;c)}e^{i\alpha_{c_+}(X)}\right], \qquad DF_L(0) = LQ_L^{\mathrm{st}}, \tag{1}$$

where

$$\phi(X;c) = \Phi^{\flat}(1,X;c), \qquad \alpha_y(X) = L^{-d}\sum_{x\in B(y)}\Phi^{\mathfrak{g}}(1,X;y,x).$$

This identity includes the endpoint corrections; it is not the uncorrected rooted logarithm alone. These source matches are checked at the deposited-transcription level. The original CMP98 full paper has not been independently reverified in this calculation.

For fixed L, the products, means and logarithms in (1) give a finite-dimensional local holomorphic map near zero. Let $D_L(c)$ be its finite set of input bonds and set

$$m_L = \sup_c |D_L(c)|, \qquad N_L = \sup_b \#\{c : b \in D_L(c)\}.$$

All translated local maps share the same constants. Write $S_L(X) = F_L(X) - LQ_L^{\mathrm{st}}X$. Since $S_L(0) = DS_L(0) = 0$, bounding the second derivative on a smaller closed complex polydisc gives constants $r_0, C_0, K_0 > 0$ such that

$$|S_L(X;c)| \le C_0 \|X\|_\infty^2, \qquad \|\partial_b S_L(X;c)\| \le K_0 \|X\|_\infty \quad (b \in D_L(c)), \tag{2}$$

whenever $\|X\|_\infty < r_0$. The derivative is zero outside $D_L(c)$. On short tori, use the lifted local pattern with multiplicities. No constant uniform as $L \to \infty$ is required.

1.2 Physical scaling and the one-step differential

At input spacing h and output spacing Lh, define

$$\mathcal{F}_h(A) = (Lh)^{-1}F_L(hA) = Q_hA + R_h(A), \qquad R_h(A) = (Lh)^{-1}S_L(hA). \tag{3}$$

Put

$$C = C_0/L, \qquad D_\infty = m_L K_0/L, \qquad D_2 = K_0\sqrt{m_L N_L}\, L^{d/2-1}.$$

For $h\|A\|_\infty < r_0$, one has

$$\|R_h(A)\|_\infty \le Ch\|A\|_\infty^2, \tag{4}$$

$$\|DR_h(A)\|_{\infty\to\infty} \le D_\infty h\|A\|_\infty, \qquad \|DR_h(A)\|_{2,h\to 2,Lh} \le D_2 h\|A\|_\infty. \tag{5}$$

Indeed each derivative matrix entry is at most $K_0h\|A\|_\infty/L$. Its row sum is at most $m_LK_0h\|A\|_\infty/L$, and its column sum in the physical adjoint normalization is at most $L^dN_LK_0h\|A\|_\infty/L$. Their geometric mean proves the Hilbert bound. This is a measure-normalized Schur estimate, with no factor from the number of fine sites in a final unit cell.

The flat average is

$$(Q_hA)_\mu(y) = L^{-d-1}\sum_{x\in B(y)}\sum_{r=0}^{L-1} A_\mu(x + rhe_\mu).$$

Its coefficients are nonnegative, its row sum is one and its unweighted column sum is L^{-d}: a given fine bond has exactly L choices of the string offset. The physical adjoint supplies the factor L^d. Thus Q_h and its physical adjoint have row sums one; in particular Q_h contracts both the bounded-field and physical Hilbert norms. Multiplicity counting gives the same assertion on short compatible tori.

1.3 The complete nonlinear iterate on a fixed complex ball

Set $h_t = L^{t-j}$ for $0 \le t \le j$ and $S = (L-1)^{-1}$. Let

$$A_0 = A, \qquad A_{t+1} = \mathcal{F}_{h_t}(A_t), \qquad \mathcal{F}^{(j)}(A) = A_j, \qquad Q_{r:s} = Q_{s-1}\cdots Q_r.$$

An empty product is the identity. Then $Q = Q_{0:j} = Q_M^{\mathrm{st}}$.

Theorem 1.1 (Uniform nonlinear ball)**.** *Choose*

$$0 < \rho < \min\{r_0/4,\, (4CS)^{-1}\}, \tag{6}$$

omitting the second restriction when $C = 0$. The complete corrected iterate is holomorphic on $\|A\|_\infty < \rho$, uniformly in refinement and volume. If $a = \|A\|_\infty$, then

$$\|A_t\|_\infty \le \frac{a}{1 - Ca\sum_{s<t} h_s} \le 2a. \tag{7}$$

Writing $R^{(j)} = \mathcal{F}^{(j)} - Q$ and $C_R = 4CS$, one also has

$$\left\|\mathcal{F}^{(j)}(A)\right\|_\infty \le 2a, \qquad \left\|R^{(j)}(A)\right\|_\infty \le C_R a^2. \tag{8}$$

Proof. Let $u_t = a/(1 - Ca\sum_{s<t} h_s)$. The denominators are positive. If $\|A_t\|_\infty \le u_t$, the local analytic domain holds because $h_t u_t \le 2\rho < r_0$. Moreover

$$u_t + Ch_t u_t^2 \le \frac{u_t}{1 - Ch_t u_t} = u_{t+1}.$$

Induction proves both the estimates and the validity of every local evaluation, without presupposing the domain bootstrap. Each composition is therefore holomorphic with a common strict margin. The exact telescoping identity

$$R^{(j)}(A) = \sum_{t=0}^{j-1} Q_{t+1:j} R_{h_t}(A_t)$$

and the contractions of $Q_{t+1:j}$ give $\left\|R^{(j)}(A)\right\|_\infty \le 4Ca^2 \sum_t h_t \le 4CSa^2$. For $j = 0$ the map is the identity and the remainder vanishes. The same local proof applies to the bounded-field Banach space on the infinite lattice. □

1.4 Differential estimates and factorial bounds

For $p = 2, \infty$, the physical intermediate norms when $p = 2$ and the ordered chain rule give

$$\left\|D\mathcal{F}^{(j)}(A)\right\|_{p\to p} \le \prod_t (1 + 2D_p a h_t) \le e^{2D_p S a}, \tag{9}$$

$$\left\|D\mathcal{F}^{(j)}(A) - Q\right\|_{p\to p} \le e^{2D_p S a} - 1 =: \delta_p(a). \tag{10}$$

For the second line, expand the difference between the product of $Q_t + DR_{h_t}(A_t)$ and the product of Q_t, keeping their noncommutative order. The nonempty selections of remainder factors sum in norm to at most $\prod_t (1 + 2D_p a h_t) - 1$. In particular no nonflat contraction is assumed.

The differential is holomorphic as an operator from fine physical L^2 to coarse physical L^2, with the field variable in bounded norm. For fixed j, the local translated Taylor expansions have finite incidence and converge in this operator norm; the preceding uniform bounds make the resulting estimates independent of volume. For $0 < r < \rho - \|A\|_\infty$ and $n \ge 1$, Cauchy gives

$$\left\|D^n\mathcal{F}^{(j)}(A)[K_1, \dots, K_n]\right\|_\infty \le 2\rho\, n!(e/r)^n \prod_i \|K_i\|_\infty . \tag{11}$$

For $n \ge 0$, with $n = 0$ read directly from (9),

$$\left\|D^{n+1}\mathcal{F}^{(j)}(A)[K_1, \dots, K_n, B]\right\|_{2,1} \le n!(e/r)^n e^{2D_2 S\rho} \|B\|_{2,\eta} \prod_i \|K_i\|_\infty . \tag{12}$$

For $n \ge 1$ use product-polydisc radii r/n for normalized directions, followed by $n^n \le e^n n!$. The mixed estimate has one Hilbert input; all other directions are bounded fields.

1.5 Finite range and weighted physical-cell operators

A one-step reading radius at most $r_L h$ gives a total physical reading radius at most $r_L \sum_t h_t \le r_L S$. Thus an integer unit-cell range R_*, including fixed cell-diameter margins, contains the support of both $D\mathcal{F}^{(j)}(A)$ and Q, independently of j. The sharper deposited two-block dependence I:24332–24336 is more than needed here.

For fine or coarse physical-cell Hilbert fibers use

$$\|T\|_\kappa = \max\left\{ \sup_x \sum_y e^{\kappa|x-y|_1} \|T_{xy}\| \,,\, \sup_y \sum_x e^{\kappa|x-y|_1} \|T_{xy}\| \right\}. \tag{13}$$

Let $v(R_*)$ count cells in that range and set $\chi_\kappa = v(R_*)e^{\kappa R_*}$. Every cell block is bounded by the global Hilbert operator norm, so

$$\left\| DR^{(j)}(A) \right\|_\kappa \le \chi_\kappa \delta_2(a). \tag{14}$$

All coefficient derivatives retain the same finite range. Hence the operator-valued Taylor series converge in the weighted norm too, and $A \mapsto DR^{(j)}(A)$ is holomorphic into this algebra. The analogous finite-range conversion for bounded-cell fibers uses $\delta_\infty(a)$. Periodic restrictions have the same bounds by lifted multiplicities or periodic cell distance. There is no fine dimension factor and no assertion of a special ordered insertion path.

1.6 An exact analytic chart for the nonlinear constraint

Let H be the already proved source-compatible flat minimizing extension, with

$$QH = I, \qquad \|H\|_{\infty\to\infty} \le k, \qquad \|H\|_\kappa \le k_\kappa. \tag{15}$$

These constants are uniform in refinement and volume. Choose $0 < r_A < \rho$ such that

$$kC_R r_A \le \tfrac{1}{4}, \qquad k\delta_\infty(r_A) \le \tfrac{1}{2}, \qquad k_\kappa \chi_\kappa \delta_2(r_A) \le \tfrac{1}{2}. \tag{16}$$

Theorem 1.2 (Uniform constraint chart)**.** *For $\xi \in \ker Q$, coarse field B, and $u = \xi + HB$ with $\|u\|_\infty < r_A/2$, the equation*

$$A = u - HR^{(j)}(A) \tag{17}$$

has a unique solution in $\|A\|_\infty \le r_A$. This defines a holomorphic map $A = \mathcal{A}_j(\xi, B)$ satisfying

$$\mathcal{F}^{(j)}(\mathcal{A}_j(\xi,B)) = B, \qquad (I - HQ)\mathcal{A}_j(\xi,B) = \xi, \tag{18}$$

$$\|\mathcal{A}_j(\xi,B) - \xi - HB\|_\infty \le 4kC_R \|\xi + HB\|_\infty^2 . \tag{19}$$

Every field with $\|A\|_\infty < r_A/3$ belongs to this chart. Its ordinary-coordinate constraint differential has a holomorphic right inverse with uniform exponential physical-cell locality.

Proof. The right side of (17) maps the closed radius r_A ball into itself: its nonlinear term has norm at most $kC_R r_A^2 \le r_A/4$. Its Lipschitz constant is at most one half by (10) and (16). The contraction theorem gives uniqueness and existence. Uniform iteration on smaller parameter balls gives holomorphic dependence.

One precise parameter norm is $\|(\xi,B)\|_{\mathrm{par}} = \|\xi + HB\|_\infty$. It is a norm because $Qu = B$, and it is uniformly equivalent to the usual direct-sum norm:

$$\|B\|_\infty \le \|u\|_\infty , \qquad \|\xi\|_\infty \le (1+k)\|u\|_\infty , \qquad \|u\|_\infty \le \|\xi\|_\infty + k\|B\|_\infty .$$

Apply Q and $I - HQ$ to (17) to obtain (18). Also $\|A\|_\infty \le 2\|u\|_\infty$, since $\left\|HR^{(j)}(A)\right\|_\infty \le \frac{1}{2}\|A\|_\infty$ on this ball. This proves (19). Conversely set $\xi = (I - HQ)A$, $B = \mathcal{F}^{(j)}(A)$. Then $u = A + HR^{(j)}(A)$ has norm below $r_A/2$ whenever $\|A\|_\infty < r_A/3$, so uniqueness identifies its fixed point with A.

At a chart point set $Q_A = D\mathcal{F}^{(j)}(A) = Q + DR^{(j)}(A)$. The ordered Neumann series give

$$H_A = (I + HDR^{(j)}(A))^{-1}H = H(I + DR^{(j)}(A)H)^{-1}, \qquad Q_A H_A = I. \tag{20}$$

In the weighted algebra the same series yield

$$\|H_A\|_\kappa \le 2k_\kappa, \qquad \left\|(I + HDR^{(j)}(A))^{-1}\right\|_\kappa \le 2. \tag{21}$$

These are respectively the B derivative of the chart and its ξ derivative restricted to $\ker Q$. The global bounded-field norms have the corresponding bounds with k. Holomorphy of the weighted inverse and Cauchy on smaller parameter balls give factorial higher derivatives as well. □

If ξ additionally satisfies the fixed flat gauge equation $RD^*\xi = 0$, the identity $RD^*H = 0$ and (17) give $RD^*A = 0$. On finite regulators this is immediate by continuity. On the infinite lattice it holds on the domain of the fixed gauge operator, or its bounded-cell extension; no new uniform bounded-field estimate for the order-zero gauge projection is asserted.

1.7 Scope of the result

This proves a refinement-uniform complex ball for the complete corrected nonlinear iterate, its physical Hilbert differential bounds and an exact analytic straightening of its constraint along the flat minimizing-extension range. The result is internally audited, not externally certified.

The coordinate is the fixed flat-reference one in (1). At nonzero fields its ordinary derivative has not here been equated with a source operator defined through left multiplication, new background-dependent endpoint corrections or a moving gauge. That identification requires the actual coordinate Jacobians and reference-background transformation. The selected chart field is not by itself an action minimizer. Notebook 14 supplies the reference-coordinate correction and a nonflat energy inverse. Notebook 15 subsequently solves the nonlinear stationarity equation in this chart, proves a strict fixed-slice minimum, and transfers it by actual gauges to the ordinary homogeneous constraint. Those later results remain restricted to a small global physical $C^{1,\alpha}$ ball. Retained or mixed-scale geometry and an interacting continuum limit remain separate tasks.

References

[1] Pinned Yang–Mills-LLM-experiment account, commit `e392ef45013a581664bab76b908671 70b30fe10e`. The corrected nonlinear composition is Raw I:24320–24325; its coordinate and flat differential are I:24344–24369; the endpoint-correction bridge is I:47076–47087.

[2] Companion research proofs: Notebook 08, the source-compatible flat minimizing extension; Notebook 11, the full flat constraint-current family; Notebook 12, pointwise regularity on the flat current domain. The original CMP98 full paper has not been independently reverified.